Perfect 👍 I'll explain your **Lecture 1 PDF (Introduction to Software Engineering, Ethics & SDLC)** in **simple words** so you can understand it clearly for your exam.

📝 **Lecture 1 – Explanation**

1. What is Software Engineering (SE)?

- It is **not just coding**.
- SE = using **engineering principles** to build software that is:
  - Reliable
  - Efficient
  - Easy to maintain
- Focuses on **structured process + quality at every step**.

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2. Why SE is Important

- Saves **time & money** in the long run.
- Protects **users** (e.g., banking apps, hospital systems, flight control).
- Avoids **big failures**.

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3. SE vs Programming

- **Programming** = just writing code.
- **SE** = whole process (analysis → design → coding → testing → deployment → maintenance).
- SE is **teamwork + structured**.
- Goal: **working, reliable, user-approved system** (not just working code).

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4. The Software Crisis (1960s–1990s)

- Many projects failed due to **over budget, missed deadlines, crashes**.

- Examples:
    - Therac-25 → patients overdosed (poor testing).
    - Ariane 5 Rocket → exploded (bad reuse of code).
    - Mariner 1 → missing hyphen destroyed rocket.
      👉 Lesson: We need **SDLC**.

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5. SDLC (Software Development Life Cycle)

- Like a **blueprint** for software.
- Ensures planning, quality, and timely delivery.
- Phases:
    1. **Feasibility Study** – Can/should we build it?
    2. **Requirements Gathering** – What does the user want?
    3. **System Design** – Blueprint (UI, database, architecture).
    4. **Implementation** – Coding (module by module).
    5. **Testing** – Find & fix errors.
    6. **Deployment** – Install & release system.
    7. **Maintenance** – Fix bugs, add features, updates.

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6. Feasibility Study

- Done **before development starts**.
- Checks:
    1. Technical – Do we have the tech skills/hardware?
    2. Economic – Is it worth the cost?
    3. Legal – Any law issues? (e.g., GDPR).
    4. Operational – Will people actually use it?
    5. Schedule – Can we finish on time?

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7. Requirements

- **Functional** → what system should do. (e.g., “Users can log in”).
- **Non-functional** → how system should behave. (e.g., “Loads in 3 seconds”).
- **Constraints** → limits. (e.g., “Runs only on Android”).

Gathering techniques: interviews, questionnaires, observation, workshops, prototyping, brainstorming.

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8. Stakeholders

- **End users** → directly use it (students/customers).
- **Internal users** → staff, admins.
- **External partners** → suppliers, payment gateways.
- **Regulatory bodies** → govt., GDPR, education ministry.
- **Project sponsor** → who funds it.

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9. System Design

- Translate requirements into **blueprints**:
    - UI design (screens, flow)
    - Database design (tables, relationships)
    - System architecture (frontend, backend, APIs).

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10. Implementation (Coding)

- Developers write code using **standards & tools**.
- Build modules (login, registration, reports).
- **Unit testing** = test each small part separately.

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11. Testing

- Types:
    1. Unit → test small parts.
    2. Integration → modules work together.
    3. System → full system works.
    4. Acceptance → client approves it.

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12. Deployment

- Install system in **real environment**.
- May start with **pilot release**.
- Train users + give manuals/help desk.

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13. Maintenance

- Starts **after release**.
- Types:
    1. Corrective → fix bugs.
    2. Adaptive → adjust to new environment (e.g., new Android).
    3. Perfective → add new features / improve performance.

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14. Ethics in Software Engineering

- SE is not just coding → must be **responsible & professional**.
- Key values:
    - **Responsibility** (own your work).
    - **Transparency** (be honest).
    - **Privacy** (protect user data).
    - **Fairness** (avoid bias).

Codes of Ethics:

- **IEEE Code** (hardware/electrical focus).
- **ACM Code** (computing focus).
- **IEEE/ACM Joint Code** (software engineers, 8 principles).
- Others: BCS (UK), ACS (Australia), IFIP (global).

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15. Example Ethical Dilemma

👉 If your manager says "Release software with bugs"
As an engineer: **Follow ethics → don't release unsafe product.**

✅ **Summary for Exam**

1. **SE = Engineering approach to software**.
2. **SE vs Programming → SE = full lifecycle, teamwork, structured**.
3. **Software Crisis → failures → SDLC created**.
4. **SDLC Phases: Feasibility → Requirements → Design → Implementation → Testing → Deployment → Maintenance**.
5. **Requirements → Functional, Non-functional, Constraints**.
6. **Stakeholders → Users, staff, sponsors, regulators**.
7. **Maintenance → Corrective, Adaptive, Perfective**.
8. **Ethics → IEEE, ACM, fairness, privacy, responsibility**.

Do you want me to also **make sample exam-style questions (MCQs + short notes)** from this PDF for your practice?

Great 👍 I'll create **50 MCQs with answers** based on your **Lecture 1: Software Engineering, Ethics & SDLC** PDF.

📘 **50 MCQs – Software Engineering (Lecture 1)**

Part A: Basics of Software Engineering

1. Software Engineering mainly focuses on:
   a) Only coding
   b) Engineering principles for software development ✅
   c) Hardware design
   d) Writing algorithms

2. Which of the following is **NOT** a characteristic of software engineering?
   a) Reliable
   b) Easy to maintain
   c) Random and unstructured ✅
   d) Efficient

3. Why does Software Engineering matter?
   a) To increase failures
   b) To protect users and save costs ✅
   c) To ignore user needs
   d) To reduce teamwork

4. A real-world example where SE is critical:
   a) Social media memes
   b) Flight control software ✅
   c) Small calculator program
   d) Notepad

5. Which is true about Software Engineering?
   a) Same as programming
   b) Larger scope than programming ✅
   c) Narrower scope than programming
   d) Only for prototypes

Part B: SE vs Programming

6. SE is different from programming because it:
   a) Is only debugging
   b) Covers complete lifecycle ✅
   c) Is less structured
   d) Requires no teamwork

7. End goal of programming is:
   a) Reliable system
   b) Working code ✅
   c) Complete lifecycle
   d) User-validated software

8. Which one is more team-oriented?
   a) Programming
   b) Software Engineering ✅

9. SE processes are:
   a) Informal
   b) Structured and well-defined ✅
   c) Random
   d) None

10. Programming is best suited for:
    a) Large systems
    b) Small scripts or prototypes ✅
    c) Banking apps
    d) Hospital systems

Part C: Software Crisis

11. The Software Crisis occurred mainly during:
    a) 1950–1960
    b) 1960–1990 ✅
    c) 2000–2010
    d) After 2015

12. Which was a famous software failure?
    a) Therac-25 ✅
    b) Microsoft Excel
    c) Facebook
    d) Instagram

13. Mariner 1 failed because of:
    a) Virus attack
    b) Missing hyphen in code ✅

c) Wrong server
d) Hardware issue

14. Ariane 5 Rocket exploded due to:
    a) Strong winds
    b) Reused code without adaptation ✅
    c) Faulty hardware
    d) Lack of fuel

15. The main lesson from Software Crisis:
    a) Avoid programming
    b) Use SDLC ✅
    c) Stop software development
    d) Ignore requirements

Part D: SDLC Basics

16. SDLC stands for:
    a) Software Development Life Cycle ✅
    b) System Data Logical Cycle
    c) Simple Design Life Code
    d) Software Documentation Life Cycle

17. Why use SDLC?
    a) To avoid planning
    b) To save time, cost, and reduce risks ✅
    c) To ignore user needs
    d) To shorten coding

18. Which is the first step of SDLC?
    a) Testing
    b) Feasibility Study ✅
    c) Maintenance
    d) Deployment

19. SDLC is similar to:
    a) Building a house without blueprint
    b) Building a house with a blueprint ✅

c) Random drawing
d) Quick notes

20. How many main phases in SDLC (as per lecture)?
    a) 4
    b) 5
    c) 6
    d) 7 ✅

Part E: Feasibility Study

21. Feasibility Study answers:
    a) “What will be coded?”
    b) “Should we build this system?” ✅
    c) “Who will test it?”
    d) “Where to deploy it?”

22. Output of a feasibility study is:
    a) User Manual
    b) Feasibility Report ✅
    c) Database Schema
    d) Test Case

23. Technical Feasibility checks:
    a) If laws are followed
    b) If required hardware/software exist ✅
    c) If deadlines are realistic
    d) If users accept it

24. Economic Feasibility means:
    a) Is it profitable/cost-effective? ✅
    b) Can it run on servers?
    c) Will users use it?
    d) Is it legal?

25. Schedule feasibility is about:
    a) Cost
    b) Time ✅

c) Hardware
d) Laws

Part F: Requirements

26. Requirements are gathered after:
    a) Design
    b) Feasibility study ✅
    c) Coding
    d) Maintenance

27. Functional requirements describe:
    a) What system should do ✅
    b) How fast system runs
    c) System limitations
    d) Who funds project

28. Non-functional requirements describe:
    a) System's behavior/performance ✅
    b) System's features
    c) Who will code it
    d) Database structure

29. A constraint means:
    a) Unlimited features
    b) A limitation ✅
    c) A new design
    d) Testing method

30. Example of functional requirement:
    a) System should allow user login ✅
    b) System should load in 3 seconds
    c) System must run on Android only
    d) All above

Part G: Stakeholders

31. Stakeholder means:
    a) Only client

b) Anyone affected/interested in system ✅
c) Only developer
d) Only tester

32. Who are end users?
    a) Developers
    b) People directly using the system ✅
    c) Sponsors
    d) Managers only

33. Regulatory bodies ensure:
    a) Design completion
    b) Legal compliance ✅
    c) Testing reports
    d) Sponsorship

34. Example of internal user:
    a) Admin staff ✅
    b) Payment gateway
    c) Suppliers
    d) Customers

35. Sponsor means:
    a) Client who funds project ✅
    b) Programmer
    c) Tester
    d) End user

Part H: SDLC Phases

36. System Design phase includes:
    a) Database schema ✅
    b) Coding
    c) Testing
    d) Manuals

37. Implementation phase is also called:
    a) Coding ✅
    b) Maintenance

c) Feasibility
d) Testing

38. Unit testing checks:
    a) Whole system
    b) Each small module ✅
    c) Legal issues
    d) Stakeholder needs

39. Deployment phase includes:
    a) Installing software in real environment ✅
    b) Writing requirements
    c) Debugging
    d) Brainstorming

40. Which is NOT a type of maintenance?
    a) Corrective
    b) Adaptive
    c) Perfective
    d) Aggressive ✅

Part I: Ethics

41. Ethics in SE ensures:
    a) Money only
    b) Responsibility, fairness, privacy ✅
    c) Only coding standards
    d) Ignoring clients

42. Example of ethical behavior:
    a) Sharing user data freely
    b) Protecting user privacy ✅
    c) Hiding bugs
    d) Releasing unsafe software

43. IEEE mainly focuses on:
    a) Hardware, electronics, computing ✅
    b) Only social sciences

c) Medicine
d) Economics

44. ACM focuses on:
    a) Software, computing research ✅
    b) Agriculture
    c) Astronomy
    d) Electrical wiring

45. Joint IEEE/ACM Code (1999) has:
    a) 2 principles
    b) 5 principles
    c) 8 core principles ✅
    d) 10 core principles

Part J: Ethics Codes

46. IEEE Code was last updated in:
    a) 1999
    b) 2020 ✅
    c) 2015
    d) 2018

47. ACM Code was last updated in:
    a) 1999
    b) 2018 ✅
    c) 2010
    d) 2020

48. Which code is widely recognized for software engineers?
    a) BCS
    b) IEEE/ACM Joint Code ✅
    c) IFIP
    d) ACS

49. BCS stands for:
    a) British Computer Society ✅
    b) Basic Computing Standards

c) Binary Code System
d) Business Computing Science

50. Ethical dilemma example from lecture:
    a) Release buggy software to meet deadline ✅
    b) Upgrade server
    c) Run feasibility study
    d) Gather requirements

✅ That's your **50 MCQs with answers**.

Would you like me to also make a **short-answer question bank (10–15 questions)** from the same PDF for theory exam practice?

Perfect 👍 I'll now give you the **answers for all 50 MCQs** in one place so you can practice quickly.

✅ **Answers – 50 MCQs (Lecture 1: Software Engineering)**

Part A: Basics of SE

1. b
2. c
3. b
4. b
5. b

Part B: SE vs Programming
6. b
7. b
8. b
9. b
10. b

Part C: Software Crisis
11. b
12. a

13. b
14. b
15. b

Part D: SDLC Basics

16. a
17. b
18. b
19. b
20. d

Part E: Feasibility Study

21. b
22. b
23. b
24. a
25. b

Part F: Requirements

26. b
27. a
28. a
29. b
30. a

Part G: Stakeholders

31. b
32. b
33. b
34. a
35. a

Part H: SDLC Phases

36. a
37. a
38. b
39. a
40. d

Part I: Ethics

41. b

42. b
43. a
44. a
45. c

Part J: Ethics Codes

46. b
47. b
48. b
49. a
50. a

✅ Now you have both the **MCQs + answers**.
Do you also want me to make a **revision cheat sheet (1-page summary of key points)** so you can read it just before exam?